

Extracted Reference Notes for the CS240 Project

Communication-Constrained Distributed Influence Maximization

Prepared from the seven PDFs in `references/`

1 Project Context

The course project studies **communication-constrained distributed influence maximization**. The target problem is to choose a seed set $S \subseteq V$, $|S| \leq k$, that maximizes expected influence spread $\sigma(S)$ under a stochastic diffusion model, while comparing centralized greedy, lazy greedy/CELF, and a simple distributed candidate-sharing scheme. The most useful material from the references is therefore:

- the formal influence maximization model and the Independent Cascade (IC) and Linear Threshold (LT) diffusion processes;
- the monotone submodular structure of $\sigma(S)$ and the $(1 - 1/e)$ greedy approximation guarantee;
- the computational bottleneck of repeated marginal-gain estimation;
- lazy evaluation as a way to reduce repeated spread evaluations;
- distributed submodular maximization as the theoretical template for “local greedy plus global merge”;
- modern scalable systems that motivate communication budgets, compression, parallel seed selection, and reverse influence sampling.

2 One-Page Synthesis

Reference	Main idea	Direct use in our project
Nemhauser et al. 1978	Classical analysis of maximizing a non-decreasing submodular set function under a cardinality constraint. Greedy achieves $1 - \left(\frac{K-1}{K}\right)^K$, approaching $1 - 1/e$.	Provides the theorem behind our greedy guarantee. We can cite it when explaining why diminishing returns make greedy a principled approximation algorithm.
Kempe et al. 2003	Introduces influence maximization under IC/LT models; proves NP-hardness; proves $\sigma(S)$ is monotone submodular; applies greedy for a near 63% guarantee.	Defines our core problem. Provides IC process, objective $\max_{ S \leq k} \sigma(S)$, NP-hardness, submodularity, and the comparison against degree/centrality heuristics.
Leskovec et al. 2007	Shows many outbreak-detection objectives are submodular and develops CELF, a cost-effective lazy forward greedy algorithm.	Supports our lazy greedy implementation. Its priority-queue logic explains why old marginal gains can serve as upper bounds after the seed set grows.
Mirzasoleiman et al. 2013	Proposes GreeDI, a two-stage distributed protocol: partition data, run local greedy, merge candidates, run greedy again.	Main template for our distributed method. We can adapt it to graph partitions: each worker returns q local candidates, and a coordinator selects final seeds.
Wang et al. 2023	PaC-IM combines sketch compression and parallel seed selection for large-scale influence maximization. It identifies space use and sequential CELF-style seed selection as bottlenecks.	Motivates why our course implementation focuses on runtime, spread-evaluation count, and scalability. We do not need to reproduce PaC-IM, but can cite it as evidence that parallel seed selection remains important.

Reference	Main idea	Direct use in our project
Chen et al. 2024	Studies scalable distributed algorithms for size-constrained monotone submodular maximization in MapReduce and adaptive-complexity models.	Gives modern theoretical context for distributed submodular maximization. Useful for framing our simplified Greedy-style approach as a course-scale version of current research.
Barik et al. 2025	GreedyRIS combines reverse influence sampling with distributed streaming maximum coverage and communication reduction techniques.	Most relevant recent paper for communication constraints. It motivates reporting candidate communication volume mq and explaining the tradeoff between quality and communication.

3 Core Definitions and Results to Reuse

3.1 Influence Maximization

Given a graph $G = (V, E)$, edge propagation probabilities, a diffusion model M , and a seed budget k , influence maximization asks for

$$\max_{S \subseteq V, |S| \leq k} \sigma(S),$$

where $\sigma(S)$ is the expected number of activated vertices when the seed set S is initially active.

For our project, the **Independent Cascade model** is the main diffusion model. When a node u first becomes active, it gets one independent chance to activate each currently inactive out-neighbor v with probability p_{uv} . The process stops when no further activation occurs. A Monte Carlo estimator can approximate $\sigma(S)$ by averaging the final active-set size over R simulation runs.

3.2 Submodularity

A set function $f : 2^V \rightarrow \mathbb{R}$ is submodular if for any $A \subseteq B \subseteq V$ and $v \notin B$,

$$f(A \cup \{v\}) - f(A) \geq f(B \cup \{v\}) - f(B).$$

This is the diminishing-returns property. For influence maximization under IC and LT, $\sigma(S)$ is monotone and submodular. This directly justifies greedy seed selection:

$$v^* = \arg \max_{v \in V \setminus S} (\sigma(S \cup \{v\}) - \sigma(S)).$$

3.3 Greedy Approximation Guarantee

Nemhauser, Wolsey, and Fisher prove that for a nonnegative, monotone submodular function under a cardinality constraint, the greedy algorithm returns a set S with

$$f(S) \geq (1 - 1/e)f(S^*),$$

up to the finite- k form $1 - \left(\frac{k-1}{k}\right)^k$. Kempe et al. apply this theorem to influence maximization after proving that the influence spread function is monotone submodular. This is the main theory bridge in our report: modern network diffusion becomes a classical submodular maximization problem.

4 Paper-by-Paper Extraction

4.1 Nemhauser, Wolsey, and Fisher (1978)

Helpful content.

- The paper studies maximizing a submodular set function with a cardinality constraint, exactly the abstract optimization form behind influence maximization.
- It treats submodularity as a discrete analogue of concavity: marginal benefit decreases as the selected set grows.
- For nondecreasing $z(S)$ with $z(\emptyset) = 0$, greedy obtains a worst-case ratio approaching $(e - 1)/e = 1 - 1/e$.

How to use it. In the theory section, state that the influence objective is an instance of monotone submodular maximization under a cardinality constraint. Then cite Nemhauser et al. for the generic greedy guarantee before citing Kempe et al. for the influence-specific proof of submodularity.

4.2 Kempe, Kleinberg, and Tardos (2003)

Helpful content.

- The paper formulates the viral-marketing seed-selection problem as influence maximization: choose k initial adopters to maximize expected final adoption.
- It defines two classical diffusion models, IC and LT. Our project should use IC as the primary implementation target because it is easy to simulate and matches the proposal.
- Influence maximization is NP-hard under IC and LT, so exact optimization is not expected in a course implementation.
- The expected influence spread $\sigma(S)$ is monotone and submodular under IC and LT. Therefore greedy has a $(1 - 1/e - \epsilon)$ -style approximation when spread values are estimated sufficiently accurately.
- Experiments compare greedy against degree and centrality heuristics and show that heuristics can waste seeds on overlapping neighborhoods. This supports our baselines: random, degree, and PageRank/centrality.

Implementation takeaway. Use Monte Carlo IC simulation for $\hat{\sigma}(S)$. The naive greedy cost is high because for each of k rounds, many candidate nodes need marginal-gain estimates. This motivates CELF and distributed candidate filtering.

4.3 Leskovec et al. (2007)

Helpful content.

- The paper studies outbreak detection but uses the same submodular optimization structure: select a limited set of nodes to maximize a reward function with diminishing returns.
- It introduces CELF, a lazy greedy method using a priority queue of cached marginal gains.
- The key observation is that, by submodularity, a previously computed marginal gain is an upper bound after the selected set grows. A node only needs reevaluation when it reaches the top of the queue.
- CELF preserves the greedy choice rule but can dramatically reduce the number of expensive objective evaluations.

Implementation takeaway. Maintain entries $(v, \hat{\Delta}_v, t_v)$, where t_v records the seed-set size when $\hat{\Delta}_v$ was last computed. Pop the largest cached gain. If it is current, select it; otherwise recompute its marginal gain under the current seed set and push it back. Report both runtime and the number of spread evaluations to show the benefit of lazy evaluation.

4.4 Mirzasoleiman et al. (2013)

Helpful content.

- The paper proposes GreeDI, a simple distributed protocol for monotone submodular maximization.
- The protocol partitions the ground set across machines, runs greedy locally, sends local selected elements to a coordinator, and runs greedy again on the union of local candidates.
- The main advantage is low communication: instead of sending the whole dataset, each worker sends only a small candidate set.
- The paper explains why distributed greedy is attractive but not identical to centralized greedy: partitions may select redundant candidates or miss nodes whose value is mainly global.

Adaptation to our project.

$$V = V_1 \cup \dots \cup V_m, \quad C_i = \text{LocalGreedy}(G_i, q), \quad C = \bigcup_{i=1}^m C_i,$$

then the coordinator returns

$$S_{\text{dist}} = \text{Greedy}(G, C, k).$$

Our experiments should vary m and q , then report

$$\rho = \frac{\sigma(S_{\text{dist}})}{\sigma(S_{\text{cent}})}$$

and the communication proxy mq .

4.5 Wang, Ding, Gu, and Sun (2023)

Helpful content.

- The paper identifies two bottlenecks in large-scale influence maximization: space-heavy sketches for marginal-gain estimation and sequential seed selection.
- PaC-IM combines sketch compression with parallel seed-selection data structures. It is far beyond the scope of our reproduction, but it gives strong motivation for measuring runtime and memory/scalability.
- The paper distinguishes Monte Carlo simulation from sketch-based methods: sketches reuse sampled graphs, often needing fewer samples than independent simulation per evaluation.
- It notes that CELF-style laziness helps reduce evaluations but can be inherently sequential, which motivates parallel or distributed variants.

How to use it. Cite this paper in the related-work section as evidence that scalable influence maximization is still an active systems problem. In our scope section, explicitly say we implement Monte Carlo greedy/CELF and a simple distributed greedy, while treating PaC-IM as advanced context.

4.6 Chen, Dey, and Kuhnle (2024)

Helpful content.

- The paper studies size-constrained monotone submodular maximization in MapReduce and adaptive-complexity models.
- It frames standard greedy as accurate but hard to scale because it needs centralized access and many adaptive rounds.
- It connects distributed algorithms with parallelizable low-adaptivity algorithms, which is useful for explaining why distributed influence maximization is not only about graph partitioning but also about reducing sequential dependence.
- It provides modern context for algorithms that keep approximation guarantees while reducing communication or adaptivity.

How to use it. Use this paper to strengthen the background paragraph on distributed submodular maximization. It is less necessary for implementation details, but valuable for positioning our simplified GreeDI experiment as a course-scale instance of a broader research direction.

4.7 Barik et al. (2025)

Helpful content.

- GreediRIS targets distributed influence maximization using reverse influence sampling (RIS).
- RIS converts seed selection into a maximum- k -coverage problem over random reverse reachable sets. A seed is good if it covers many sampled reverse reachable sets.
- The paper focuses on the seed-selection bottleneck in distributed RIS systems and uses a distributed streaming max-cover approach.
- It explicitly studies communication bottlenecks and introduces truncation as a knob that trades quality for lower communication.

How to use it. This is the strongest recent support for our communication-constrained angle. We can cite it when defining q as a local candidate budget and when explaining why increasing q should improve quality but increase communication.

5 Recommended Report Integration

5.1 Stage 1: Background and Related Work

Use Kempe et al. for the application background and diffusion model. Use Nemhauser et al. for submodular maximization. Use Leskovec et al. for lazy evaluation. Use Mirzasoleiman et al., Chen et al., Wang et al., and Barik et al. to motivate distributed and scalable variants.

5.2 Stage 2: Representative Method

The representative method should be **centralized greedy with IC Monte Carlo estimation**, plus CELF as the main acceleration. The distributed extension should be a GreeDI-style method rather than a full RIS system, because it is easier to implement and directly matches the proposal.

5.3 Stage 3–4: Implementation and Experiments

Recommended algorithms:

- Random baseline.
- Degree baseline.

- PageRank or centrality baseline.
- Monte Carlo greedy.
- CELF/lazy greedy.
- Distributed local greedy plus coordinator greedy.

Recommended metrics:

- estimated influence spread $\hat{\sigma}(S)$;
- runtime;
- number of spread evaluations;
- distributed quality ratio ρ ;
- communication proxy mq ;
- sensitivity to m, q, k , and propagation probability.

6 Useful Text Snippets for the Final Report

Problem motivation. Influence maximization turns viral marketing, information diffusion, and intervention planning into a cardinality-constrained combinatorial optimization problem on graphs. Its importance comes from the fact that small seed sets can trigger large cascades, but selecting seeds purely by local centrality may waste budget on overlapping neighborhoods.

Theory bridge. Under the IC model, the expected spread function is monotone and submodular. Thus each new seed has diminishing marginal benefit as the current seed set grows. This property connects influence maximization to the classical Nemhauser-Wolsey-Fisher theorem, giving the greedy algorithm a $(1 - 1/e)$ approximation guarantee.

Why CELF. The main cost of greedy influence maximization is repeated spread estimation. CELF exploits submodularity by treating cached marginal gains as upper bounds, recomputing a node only when it may still be the best candidate. Therefore it preserves the greedy selection rule while reducing objective evaluations.

Why distributed greedy. Centralized greedy assumes full access to the graph and candidate set. In distributed settings, each worker can select local candidates and send only a small set to a coordinator. This reduces communication but may lose quality when important cross-partition influence is not represented in the candidate pool.

References

- [1] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher. An analysis of approximations for maximizing submodular set functions - I. *Mathematical Programming*, 14:265–294, 1978.
- [2] D. Kempe, J. Kleinberg, and E. Tardos. Maximizing the spread of influence through a social network. *KDD*, 2003.
- [3] J. Leskovec, A. Krause, C. Guestrin, C. Faloutsos, J. VanBriesen, and N. Glance. Cost-effective outbreak detection in networks. *KDD*, 2007.
- [4] B. Mirzasoleiman, A. Karbasi, R. Sarkar, and A. Krause. Distributed submodular maximization: identifying representative elements in massive data. *NeurIPS*, 2013.
- [5] L. Wang, X. Ding, Y. Gu, and Y. Sun. Fast and space-efficient parallel algorithms for influence maximization. *PVLDB*, 17(3):400–413, 2023.
- [6] Y. Chen, T. Dey, and A. Kuhnle. Scalable distributed algorithms for size-constrained submodular maximization in the MapReduce and adaptive complexity models. *Journal of Artificial Intelligence Research*, 80:1575–1622, 2024.
- [7] R. Barik, W. Cappa, S. M. Ferdous, M. Minutoli, M. Halappanavar, and A. Kalyanaraman. GreediRIS: Scalable influence maximization using distributed streaming maximum cover. *Journal of Parallel and Distributed Computing*, 198:105037, 2025.